

46 Формула конечных приращений

Найдем формулу для приращений функции.

$$y = f(x), \Delta y = f(x + \Delta x) - f(x) = f'(c) \Delta x = f'(x + \theta \Delta x) \Delta x$$

$$\theta \in (0, 1)$$

$$\text{Пусть } z = f(x, y) \quad \Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$$

$$\Delta F = F(1) - F(0) = F'_t(c) \Delta t$$

$$\text{Пусть } F(t) = f(x_0 + t \Delta x, y_0 + t \Delta y) = f(x, y)$$

$$\Delta F = F(1) - F(0) = F'_t(c) \Delta t \quad \text{или}$$

$$x = x_0 + t \Delta x$$

$$y = y_0 + t \Delta y$$

$$0 \leq t \leq 1$$

$$\equiv \left(F'_x(x_0 + t\Delta x, y_0 + t\Delta y) \cdot \frac{dx}{dt} + F'_y(x_0 + t\Delta x, y_0 + t\Delta y) \frac{dy}{dt} \right) \Big|_{t=c}$$

$$\Delta f = (f'_x(x_0 + \Theta\Delta x, y_0 + \Theta\Delta y)\Delta x + f'_y(x_0 + \Theta\Delta x, y_0 + \Theta\Delta y)\Delta y)$$

$$\frac{dx}{dt} = \frac{d(x_0 + t\Delta x)}{dt} = \Delta x; \quad \frac{dy}{dt} = \frac{d(y_0 + t\Delta y)}{dt} = \Delta y$$